

Cleaning the dataset to ensure consistency and usability

Old Dataset

Customer_ID	Transaction_Date	Product_Category	Amount_Spent	Customer_Region
C001	15.7.2023	Electronics	\$120.50	North
C002	15.7.2023	Elec.	€95.00	East
C003	07/16/2023	electronic	105.75	West
C004	17.7.2023	Fashion	89.99	North
C005	17.7.2023	Home & Garden	\$	South
C006	18.7.2023	Electronics	130.00	East
C002	15.7.2023	Elec.	€95.00	East
C007	07/19/2023	Fashion	NaN	West
C008	20.7.2023	Home & Garden	\$75.25	North
C009	20.7.2023	Electronics	110.00	South

New Dataset

Customer_ID	Transaction_Date	Product_Category	Amount_Spent	Customer_Region
C001	15.7.2023	Electronics	120.50	North
C002	15.7.2023	Electronics	95.00	East
C003	16.7.2023	Electronics	105.75	West
C004	17.7.2023	Fashion	89.99	North
C005	17.7.2023	Home & Garden	0.00	South
C006	18.7.2023	Electronics	130.00	East
C007	19.7.2023	Fashion	0.00	West
C008	20.7.2023	Home & Garden	75.25	North
C009	20.7.2023	Electronics	110.00	South

There is a second time C002 with same date, category and amount spent. I am removing it, because it may mislead. Also I am fixing the date – all date to be the same way (understandable enough). In product category column, there is Electronics, but there is also Elec., electronic. I am renaming them to be with same word – Electronica. Some values are missing, I am setting them to 0. And some values are with currencies, some not. I will remove them, because I have no idea what is the currency for those I do not see.

Identify the top 3 regions by total revenue

North – $120.50 + 89.99 + 75.25 = 285.74$

South – $0.00 + 110.00 = 110.00$

West – $105.75 + 0.00 = 105.75$

East – $95.00 + 130.00 = 225.00$

Top 3 Revenue for business (client payment)

1. North – 285.74;
2. East – 225.00;
3. South – 110.00

According to the old dataset, unfortunately we do not know the currency of some of the prices and because of that I am analyzing the values without currencies!

Which product category has the highest average spend per transaction?

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C002	15.7.2023	Electronics	95.00	East
C003	16.7.2023	Electronics	105.75	West
C004	17.7.2023	Fashion	89.99	North
C005	17.7.2023	Home & Garden	0.00	South
C006	18.7.2023	Electronics	130.00	East
C007	19.7.2023	Fashion	0.00	West
C008	20.7.2023	Home & Garden	75.25	North
C009	20.7.2023	Electronics	110.00	South

I am bringing the dataset again (the new one and repaired). So, let's calculate again:

- *Electronics* – $120.50 + 95.00 + 105.75 + 130.00 + 110.00 = 561.25$ (Total)
- *Fashion* – $89.99 + 0.00 = 89.99$ (Total)
- *Home & Garden* – $0.00 + 75.25 = 75.25$ (Total)

Average spend:

Electronics – $561.25 / 5 = 112.25$ (Total); - 1st place

Fashion – $89.99 / 2 = 44.995$ (Total); - 2nd place

Home & Garden – $75.25 / 2 = 37.625$ (Total) – 3rd place

Conclusion

According to my research, I discovered the following important information:

The North part of the region has the biggest revenue, perhaps because of the most people that present there or maybe the products are the best.

The category that is getting the attention of the people mostly is the Electronics. It looks like that most people buying it and the average spend is the highest.

This analysis is created by:



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During the analysis the following platforms and programs were used:

Microsoft Word